

Computing Algebraic Spectral Sequences over PID-Type Coefficients

Kunko (Yuankun) Zou

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Abstract

This paper develops a concrete algebraic framework for computing page modules and reconstructing differential image data in multiplicative spectral sequences over principal ideal domains and closely related computational coefficient rings. The central idea is to represent each homogeneous piece as a quotient of one explicit column space by another inside an ambient free module, so that Smith normal form can be used to compute structure, solve exact linear problems, determine kernels and images, and detect torsion on later pages. After introducing the general formalism, the paper explains how absolute coordinates, quotient-module presentations, and alignment arguments combine into a recursive page-building procedure. A program-level guide over $\mathbb{Z}$ then shows how to enter a concrete spectral sequence and inspect the torsion that appears on later pages.

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1 Introduction

Spectral sequences are often introduced as a bookkeeping device for complicated homological computations [1, 2, 3, 4]. In practice, however, once one fixes a specific algebraic spectral sequence, one still needs concrete algorithms for computing the modules on each page, for determining the

effect of differentials, and for extracting structural information such as free rank and torsion. The goal of this paper is to explain one such computational framework.

The point of view taken here is purely algebraic. We begin with a commutative bigraded algebra presented by generators and relations over a principal ideal domain, and we equip each page with a differential of prescribed bidegree. The next page is obtained by taking homology. The main computational problem is therefore the repeated evaluation of kernels and images inside finitely generated modules over a PID, a standard setting in homological algebra [2, 5]. The key algebraic tool is Smith normal form [8, 9].

The central idea is simple. At each bidegree, instead of working with abstract quotient modules directly, we represent the module by two matrices whose columns lie in an ambient free module:

$$\text{module at bidegree } b \cong \text{Col}(S_b) / \text{Col}(Z_b).$$

Here S_b is a span matrix and Z_b is a relation matrix. Once both are expressed in coordinates inside a free module over the base domain, Smith normal form can be used to

1. determine the structure of the quotient,
2. decide membership in column spaces,
3. solve linear equations exactly,
4. compute kernels of differentials modulo relations, and
5. align partial differential data with the full span matrix.

This leads to a recursive procedure for building the pages $E_1, E_2, E_3, \dots$ of an algebraic spectral sequence once the relevant differential values are supplied or uniquely inferred [1, 6]. The same formalism also explains how torsion can appear naturally on later pages, even when the earlier pages look free.

There is one important distinction between the mathematics and the implementation. The abstract structural results in this paper are stated over a PID. However, the actual algorithms require exact operations such as extended gcd, remainder, and exact quotient, so the implementation is really written for what one may call *computational PID domains*, and in practice behaves most naturally over Euclidean-style domains such as $\mathbb{Z}$. (Of course, the algorithm also works over fields, but in that setting one would usually use ordinary linear algebra for efficiency considerations.) We will state the structural results over PIDs and isolate the stronger algorithmic assumptions when needed.

The paper is organized as follows. In Section 2 we recall the algebraic definition of a spectral sequence in the form used here. In Section 3 we explain how the bigraded algebras in the implementation arise from polynomial rings with assigned generator bidegrees, how homogeneous elements are represented in absolute coordinates, and how these coordinates organize the recursive page construction, deferring the discussion of differential input and inference to Section 5. Section 4 reviews Smith normal form, alignment, exact division, and the quotient-module computations used to build later pages. Section 5 explains how differential image data is supplied, inferred, and completed. Section 6 collects some additional computational details. Section 7 gives a program-level guide to the sample spectral sequence over $\mathbb{Z}$. Section 8 discusses the limitations of the present implementation, and Section 9 concludes.

2 Algebraic Spectral Sequences

This section fixes the algebraic language for the spectral sequences considered throughout the paper.

Bigraded modules and algebras

We begin by recalling the basic bigraded objects that underlie the later constructions.

Let R be a commutative ring. A *bigraded R -module* is an R -module M together with a decomposition

$$M = \bigoplus_{(p,q) \in \mathbb{Z}^2} M^{p,q}.$$

An element $x \in M^{p,q}$ is called *homogeneous of bidegree (p, q)* .

A *bigraded R -algebra* is a bigraded R -module $A = \bigoplus A^{p,q}$ equipped with an associative R -bilinear multiplication such that

$$A^{p,q} \cdot A^{p',q'} \subseteq A^{p+p', q+q'}.$$

Throughout this paper, all R -algebras are assumed commutative unless stated otherwise.

In the computational setting of this paper, the bigraded algebra is usually generated by finitely many homogeneous generators $x_1, \dots, x_n$ with prescribed bidegrees. One then works with homogeneous polynomials in those generators modulo relations.

Bigraded submodules and quotients

We next record the basic criterion that allows one to recognize when a submodule respects the ambient bigrading.

Definition 2.1. Let $M = \bigoplus_{(p,q) \in \mathbb{Z}^2} M^{p,q}$ be a bigraded R -module, and let $N \subseteq M$ be a submodule. We say that N is a bigraded submodule if

$$N = \bigoplus_{p,q} (N \cap M^{p,q}).$$

Proposition 2.2 (Bigraded submodule criterion). *Under the hypotheses of the definition, N is a bigraded submodule if and only if every $x \in N$ can be written as a finite sum of homogeneous elements of N .*

Proof. The condition is really saying that the collection $\{N \cap M^{p,q}\}$ spans N by finite sum, so we only need to establish the uniqueness of this decomposition. Indeed, if $x \in N$ has a finite decomposition

$$x = \sum_{p,q} x_{p,q}, \quad x_{p,q} \in N \cap M^{p,q},$$

this decomposition is unique because elements of M decompose uniquely into its homogeneous components. $\square$

A seemingly stronger claim is the following: if N is a bigraded submodule and we have $x \in N$ with any decomposition

$$x = \sum_{p,q} x_{p,q}, \quad x_{p,q} \in M^{p,q},$$

then $x_{p,q} \in N$. This is actually immediate because N being a bigraded submodule gives a decomposition of x into $N \cap M^{p,q}$ and this decomposition is unique in M . As a corollary, quotients by bigraded submodules inherit a bigrading.

Proposition 2.3 (Bigraded module quotients). *Let*

$$M = \bigoplus_{(p,q) \in \mathbb{Z}^2} M^{p,q}$$

be a bigraded R -module, and let $N \leq M$ be a bigraded submodule. Then M/N is a bigraded R -module with homogeneous pieces

$$(M/N)^{p,q} := (M^{p,q} + N)/N \cong M^{p,q}/(N \cap M^{p,q}).$$

where the isomorphism is the standard isomorphism theorem.

Proof. First, $(M^{p,q} + N)/N$ spans all of M/N by finite sums, because one can simply pass a decomposition in M to the quotient. It remains to prove that the sum is direct. Suppose that for some finite subset $F \subseteq \mathbb{Z}^2$ we have

$$\sum_{(p,q) \in F} (m_{p,q} + N) = N$$

with each $m_{p,q} \in M^{p,q}$. Then

$$\sum_{(p,q) \in F} m_{p,q} \in N.$$

Because N is a bigraded submodule, $m_{p,q} \in N$ by the remark following Proposition 2.2. Hence each summand satisfies

$$m_{p,q} + N = N,$$

so all terms are zero in the quotient. Therefore the above sum is direct, and we obtain

$$M/N = \bigoplus_{(p,q) \in \mathbb{Z}^2} \frac{M^{p,q} + N}{N}.$$

□

The same idea applies to ideals inside bigraded algebras.

Proposition 2.4 (Bigraded algebra quotients). *Let $A = \bigoplus A^{p,q}$ be a bigraded R -algebra, and let $B \subseteq A$ be an ideal that is also a bigraded submodule of A . Then the quotient A/B is a bigraded R -algebra with homogeneous pieces*

$$(A/B)^{p,q} = (A^{p,q} + B)/B.$$

Proof. By Proposition 2.3, A/B is already a bigraded R -module, so we only need to check that the quotient-ring multiplication is well-defined and respects bidegree. The first part is because B is an ideal of A : if $a_1 - a \in B$ and $a'_1 - a' \in B$, then

$$a_1 a'_1 - aa' = (a_1 - a)a'_1 + a(a'_1 - a') \in B.$$

Now if $a \in A^{p,q}$ and $a' \in A^{p',q'}$, then $aa' \in A^{p+p',q+q'}$, so this product lies in

$$(A^{p+p',q+q'} + B)/B.$$

Thus multiplication respects the induced bidegree, so A/B is a bigraded R -algebra. □

Differentials

We begin by fixing the basic notion of a differential in the bigraded setting.

Definition 2.5. Let $M = \bigoplus M^{p,q}$ be a bigraded R -module. A differential of bidegree (a, b) on M is an R -linear map

$$d : M \rightarrow M$$

such that

$$d(M^{p,q}) \subseteq M^{p+a, q+b}$$

for all (p, q) , and

$$d^2 = 0.$$

If A is a bigraded R -algebra, a differential d on A is called a *derivation* if for homogeneous $x, y \in A$ one has

$$d(xy) = d(x)y + x d(y).$$

Throughout this paper, we use sign-free Leibniz rule for simplicity. Actually, since we only care about the computational aspect, this sign issue does not really matter.

The implementation does not attempt to generate the full Leibniz closure of the initially supplied differential data automatically. Instead, it uses a restricted collection of inference rules, described in Section 5, and requests additional input when those rules do not determine a unique answer.

Propositions 2.2, 2.3, and 2.4 immediately control kernels, images, and quotients in this setting.

Proposition 2.6 (Homology bigrading). *Let M be a bigraded R -module with a differential d of bidegree (a, b) . Then the homology*

$$H(M, d) = \ker(d) / \operatorname{im}(d)$$

inherits a natural bigrading, with homogeneous pieces

$$H(M, d)^{p,q} = \frac{(\ker(d) \cap M^{p,q}) + \operatorname{im}(d)}{\operatorname{im}(d)} \cong \frac{\ker(d) \cap M^{p,q}}{\operatorname{im}(d) \cap M^{p,q}}.$$

Proof. We first show that both $\ker(d)$ and $\operatorname{im}(d)$ are bigraded submodules of M .

Let $x \in \ker(d)$, and write its homogeneous decomposition as

$$x = \sum_{p,q} x_{p,q}, \quad x_{p,q} \in M^{p,q}.$$

Then

$$0 = d(x) = \sum_{p,q} d(x_{p,q}),$$

and the translation $(p, q) \mapsto (p + a, q + b)$ is injective on $\mathbb{Z}^2$. Hence the summands $d(x_{p,q})$ lie in distinct bidegrees $M^{p+a, q+b}$, so every $d(x_{p,q})$ vanishes. Therefore each $x_{p,q} \in \ker(d)$, and the criterion for bigraded submodules shows that $\ker(d)$ is bigraded.

Now let $y \in \operatorname{im}(d)$. Choose $x \in M$ with $y = d(x)$, and write

$$x = \sum_{p,q} x_{p,q}, \quad x_{p,q} \in M^{p,q}.$$

Then

$$y = \sum_{p,q} d(x_{p,q}),$$

and each $d(x_{p,q})$ is homogeneous of bidegree $(p+a, q+b)$. Thus every homogeneous component of y again lies in $\text{im}(d)$, so the same criterion shows that $\text{im}(d)$ is bigraded.

Since $d^2 = 0$, one has $\text{im}(d) \subseteq \ker(d)$. Applying Proposition 2.3 to the bigraded module $\ker(d)$ and its bigraded submodule $\text{im}(d)$, we obtain

$$H(M, d)^{p,q} = \frac{(\ker(d) \cap M^{p,q}) + \text{im}(d)}{\text{im}(d)}.$$

Since $\text{im}(d) \subseteq \ker(d)$, the standard isomorphism theorem identifies this with

$$\frac{\ker(d) \cap M^{p,q}}{\text{im}(d) \cap M^{p,q}}.$$

□

The multiplicative structure is equally well behaved.

Proposition 2.7 (Homology algebra). *Let A be a bigraded R -algebra with a differential d that is a derivation. Then multiplication descends to the homology $H(A, d)$, so $H(A, d)$ is again a bigraded R -algebra.*

Proof. By Proposition 2.6, both $\ker(d)$ and $\text{im}(d)$ are bigraded submodules of A . We next check the algebraic conditions needed to apply Proposition 2.4.

If $x, y \in \ker(d)$, then

$$d(xy) = d(x)y + x d(y) = 0,$$

so $\ker(d)$ is a subalgebra of A .

Now let $z \in \ker(d)$ and let $d(u) \in \text{im}(d)$. Then

$$d(u)z = d(uz) - u d(z) = d(uz), \quad z d(u) = d(zu) - d(z)u = d(zu),$$

and both products lie in $\text{im}(d)$. Thus $\text{im}(d)$ is an ideal in the subalgebra $\ker(d)$. Since it is already a bigraded submodule, it is a bigraded ideal.

Therefore Proposition 2.4 applies to

$$H(A, d) = \ker(d) / \text{im}(d),$$

and shows that multiplication descends to homology and that the quotient is again a bigraded R -algebra. □

Algebraic spectral sequences

With these preliminaries in place, we can write down the version of a spectral sequence used throughout the paper.

Definition 2.8. An algebraic spectral sequence over R in the sense of this paper consists of:

1. bigraded R -algebras $E_r = \bigoplus E_r^{p,q}$ for $r \geq 1$,

2. differentials

$$d_r : E_r^{p,q} \rightarrow E_r^{p+\alpha_r, q+\beta_r},$$

usually of the form $(\alpha_r, \beta_r) = (r, 1 - r)$,

3. isomorphisms of bigraded R -algebras

$$E_{r+1} \cong H(E_r, d_r).$$

One remark we need to make here is that we can choose

$$E_{r+1}^{p,q} = \frac{\ker(d_r) \cap E_r^{p,q}}{\text{im}(d_r) \cap E_r^{p,q}}.$$

Then $E_{r+1}^{p,q}$ is obtained from $E_r^{p,q}$ by restricting to the kernel and then quotienting by the incoming image. It is therefore natural to choose, for each bidegree b , a fixed ambient free module F^b in which all later page representatives at b live; later pages are produced by shrinking the span and enlarging the relation submodule inside that same F^b . We make this precise in the next section.

3 Formulation of the computation problem

The central computational problem is straightforward: given the first page and information about differentials at each page, we want to compute the ranks and structural data of each bidegree at some later page, up to isomorphism. Three practical questions then arise:

1. How should we represent the modules across pages?
2. How should we represent elements of the bigraded algebra?
3. How is differential information supplied and partially inferred?

The first two questions are addressed in this section. The third depends on the exact-division and alignment machinery developed later, so we defer it to Section 5.

Page Recursion

To make the abstract modules computable, we need to choose a way of representing them. As a classical result, every module over a ring with identity can be represented as a quotient of a free module. However, for reasons that become important later, we want a slightly more flexible setup. For each bidegree $b = (p, q)$, we will work in an ambient finite-dimensional free module F^b , and on page 1 we will set

$$E_1^b = T_1^b / N_1^b,$$

where T_1^b and N_1^b are both submodules of F^b . Also, since d_1 is homogeneous of bidegree δ , we can view it as R -linear maps

$$E_1^{b-\delta} \xrightarrow{d_1} E_1^b \xrightarrow{d_1} E_1^{b+\delta}.$$

where δ is the bidegree of d . This setup is still quite general: it covers any first page whose homogeneous components admit finite-rank free presentations. The goal is to preserve this style of presentation on later pages as well, and for that we need one elementary quotient calculation.

Proposition 3.1. Suppose that we have the following commutative diagram of R -modules,

$$\begin{array}{ccccc} S' & \xrightarrow{\bar{f}} & S & \xrightarrow{\bar{g}} & S'' \\ \downarrow \pi' & & \downarrow \pi & & \downarrow \pi'' \\ S'/K' & \xrightarrow{f} & S/K & \xrightarrow{g} & S''/K'' \end{array}$$

where all vertical arrows are quotient maps and $g \circ f = 0$. Then

$$\ker(g)/\operatorname{im}(f) \cong (\bar{g})^{-1}(K'')/(\bar{f}(S') + K).$$

Proof. First, $g \circ f = 0$, so $(\bar{g} \circ \bar{f})(S') \subseteq K''$, in other words, $\bar{f}(S') \subseteq \bar{g}^{-1}(K'')$. Also, we have $\bar{g}(K) \subseteq K''$ for g to be well-defined. Combining them, we have

$$\bar{f}(S') + K \subseteq \bar{g}^{-1}(K'').$$

Now,

$$\operatorname{im}(f) = \operatorname{im}(f \circ \pi') = \operatorname{im}(\pi \circ \bar{f}) = (\operatorname{im}(\bar{f}) + K)/K.$$

Also, for $x \in S$

$$x + K \in \ker(g) \iff \bar{g}(x) + K'' = g(x + K) = K'' \iff \bar{g}(x) \in K''.$$

Therefore,

$$\ker(g) = \{x + K : x \in \bar{g}^{-1}(K'')\} = (\bar{g}^{-1}(K'') + K)/K = \bar{g}^{-1}(K'')/K.$$

where the last equality follows from $K \subseteq \bar{g}^{-1}(K'')$. Then, the third isomorphism gives

$$\frac{\ker(g)}{\operatorname{im}(f)} \cong \frac{\bar{g}^{-1}(K'')/K}{(\bar{f}(S') + K)/K} \cong \frac{\bar{g}^{-1}(K'')}{\bar{f}(S') + K}.$$

□

Proposition 3.2. Suppose that on page r , we have quotient presentations $E_r^b = T_r^b/N_r^b$, where $T_r^b \subseteq F^b$, $N_r^b \subseteq F^b$, and there are maps $\bar{d}_r$ satisfying

$$\begin{array}{ccccc} T_r^{b-\delta} & \xrightarrow{\bar{d}_r} & T_r^b & \xrightarrow{\bar{d}_r} & T_r^{b+\delta} \\ \downarrow & & \downarrow & & \downarrow \\ E_r^{b-\delta} & \xrightarrow{d_r} & E_r^b & \xrightarrow{d_r} & E_r^{b+\delta} \end{array}$$

and

$$\bar{d}_r(T_r^b) \subseteq T_r^{b+\delta}, \quad \bar{d}_r(N_r^b) \subseteq N_r^{b+\delta}.$$

Then, we can choose $N_{r+1}^b \subseteq T_{r+1}^b \subseteq F^b$ and set

$$E_{r+1}^b = T_{r+1}^b/N_{r+1}^b.$$

Proof. Apply Proposition 3.1 with $f = d_r$ and $g = d_r$. The hypothesis $g \circ f = 0$ is exactly the condition $d_r^2 = 0$. The proposition gives a quotient presentation, so in particular the relation submodule below is contained in the span submodule below. This yields

$$T_{r+1}^b = (\bar{d}_r)^{-1}(N_r^{b+\delta}), \quad N_{r+1}^b = \bar{d}_r(T_r^{b-\delta}) + N_r^b.$$

□

For convenience, we will also use the notation $T_r^b = \operatorname{Col}(S_b)$ and $N_r^b = \operatorname{Col}(Z_b)$ when the page number r is clear from the context. We call S_b the span matrix at bidegree b and Z_b the relation matrix.

Initial relations

Of course, on the first page each bidegree could be given its own quotient presentation. In practice, however, we do not want to construct those presentations independently. Instead, we want to start with an algebra whose homogeneous pieces are free and then mod out by a bigraded ideal. For this reason, we establish that an ideal generated by homogeneous elements is automatically a bigraded ideal.

Proposition 3.3 (Homogeneous ideals give bigraded quotients). *Let A be a commutative bigraded algebra over R , and let $I \subseteq A$ be an ideal generated by homogeneous elements. Then I is a bigraded ideal of A . Consequently A/I is a bigraded R -algebra.*

Proof. Take any element $f \in I$. Since I is generated by homogeneous elements, one can write

$$f = \sum_{j=1}^N h_j r_j,$$

where each r_j is homogeneous and each $h_j \in A$.

Now decompose each coefficient h_j into its homogeneous pieces:

$$h_j = \sum_{p,q} h_j^{p,q}, \quad h_j^{p,q} \in A^{p,q}.$$

where only finitely many $h_j^{p,q}$ are non-zero. Reorganizing gives

$$f = \sum_{p,q} \sum_{j=1}^N h_j^{p,q} r_j.$$

Because each r_j is homogeneous, every product $h_j^{p,q} r_j$ is again homogeneous. Moreover each such product lies in I , since I is an ideal. Therefore f is a sum of homogeneous elements of I . By Proposition 2.2, this shows that I is a bigraded submodule of A . Since I was already an ideal, it is a bigraded ideal. Proposition 2.4 therefore shows that A/I is a bigraded R -algebra. $\square$

In practice, we mod out a relation by specifying a homogeneous element in the bigraded algebra, and in each specific homogeneous component of the first page, we construct the local N_1^b by searching all multiples of all relations that occur.

For this reason, the computation of each module on the first page begins with:

1. the full absolute basis as columns of the span matrix,
2. all relation multiples landing in the given bidegree as columns of the relation matrix.

Bigraded Polynomial Algebra

We now turn to the second question about representing elements in the bigraded algebra. The quotient presentations are already sufficient for computing homology, but in practice the first page presentations need to be generated in a convenient way. Also, we need to keep track of multiplication in the bigraded algebra for later functionality, such as inference of differential information using the Leibniz rule.

One convenient way to get a bigraded algebra is to start from a polynomial ring over a commutative ring and then specify bidegrees on each generator. Besides conceptual simplicity, this model

is also easy for programs to manipulate. The central idea is the remark we made in the last section that modules at a fixed degree across different pages can be unified: they are just submodules and quotients of an ambient module, which will be described in terms of “absolute basis” and “absolute coordinates”.

Let R be a commutative ring and consider a polynomial algebra

$$R[x_1, \dots, x_n].$$

Assume each generator x_i is assigned a bidegree

$$|x_i| = (a_i, b_i) \in \mathbb{Z}^2.$$

It is convenient to collect these into a $2 \times n$ matrix

$$G = \begin{bmatrix} a_1 & \cdots & a_n \\ b_1 & \cdots & b_n \end{bmatrix}.$$

For an exponent vector

$$e = (e_1, \dots, e_n)^T \in \mathbb{N}^n,$$

the monomial

$$x^e := x_1^{e_1} \cdots x_n^{e_n}$$

has *absolute bidegree*

$$|x^e| = Ge.$$

The point of the preceding assignment is that it indeed defines a bigrading on the whole polynomial algebra.

Proposition 3.4. *Let $A = R[x_1, \dots, x_n]$, and assign bidegrees $|x_i| = (a_i, b_i)$ to the generators. For each $(p, q) \in \mathbb{Z}^2$, let*

$$A^{p,q} = \text{span}_R\{x^e : Ge = (p, q)\}.$$

Then

$$A = \bigoplus_{(p,q) \in \mathbb{Z}^2} A^{p,q}$$

and

$$A^{p,q} \cdot A^{p',q'} \subseteq A^{p+p', q+q'}.$$

Hence A is a bigraded R -algebra.

Proof. Every polynomial in A is a finite R -linear combination of monomials x^e . Grouping those monomials according to the value of Ge shows that every polynomial lies in $\sum_{p,q} A^{p,q}$.

This sum is direct because distinct monomials are R -linearly independent. So if a polynomial is written as a sum of pieces supported in different bidegrees, those pieces are uniquely determined.

Now let $x^e \in A^{p,q}$ and $x^f \in A^{p',q'}$. Then

$$x^e x^f = x^{e+f}$$

and

$$|x^{e+f}| = G(e+f) = Ge + Gf = (p+p', q+q').$$

So the product of two homogeneous monomials has the expected bidegree. □

Thus $A^{p,q}$ is simply the module of all polynomials all of whose monomials have the same absolute bidegree. Next, we discuss the technical conditions that ensure $A^{p,q}$ is free of finite rank.

Fix a bidegree $b \in \mathbb{Z}^2$. The set of exponent vectors of monomials of bidegree b is

$$\mathcal{E}_b = \{e \in \mathbb{N}^n : Ge = b\}.$$

If $\mathcal{E}_b$ is finite, then the corresponding monomials

$$\{x^e : e \in \mathcal{E}_b\}$$

form a distinguished basis of the homogeneous piece of bidegree b . We will call this the *absolute basis* at bidegree b .

Proposition 3.5 (Finite absolute basis). *Assume:*

1. the second coordinates b_i of the generator bidegrees are nonnegative,
2. whenever $b_i = 0$, one has $a_i > 0$,
3. the fixed bidegree $b = (p, q)$ satisfies $q \geq 0$.

Then $\mathcal{E}_b$ is finite.

Proof. Let

$$I_+ = \{i : b_i > 0\}, \quad I_0 = \{i : b_i = 0\}.$$

If $e \in \mathcal{E}_b$, then the second-coordinate equation is

$$\sum_{i \in I_+} b_i e_i = q.$$

Hence for every $i \in I_+$,

$$0 \leq e_i \leq \frac{q}{b_i},$$

so all coordinates with positive second component are bounded above.

Now consider the first-coordinate equation:

$$\sum_{i \in I_0} a_i e_i = p - \sum_{i \in I_+} a_i e_i.$$

The right-hand side is bounded above because the coordinates with $b_i > 0$ are already bounded above. Even if some of the corresponding a_i are negative, their total negative contribution is bounded. Therefore

$$\sum_{i \in I_0} a_i e_i \leq p - \sum_{i \in I_+, a_i < 0} a_i \cdot \frac{q}{b_i} =: M.$$

For $i \in I_0$, assumption (2) gives $a_i > 0$. Since

$$\sum_{i \in I_0} a_i e_i \leq M,$$

each coordinate with $b_i = 0$ also satisfies

$$0 \leq e_i \leq \frac{M}{a_i}.$$

Thus every coordinate of e is bounded above, so only finitely many exponent vectors can occur. $\square$

Proposition 3.5 explains why one can search for all monomials of a given bidegree by finite combinatorics. In practice, we search them via an almost brute-force method.

Once finiteness is known, the coordinate description is immediate.

Proposition 3.6 (Absolute basis expansion). *For any fixed bidegree b , every homogeneous polynomial of absolute bidegree b can be written uniquely as an R -linear combination of the absolute basis monomials at b .*

Proof. Distinct monomials in a polynomial ring are R -linearly independent. By construction, every homogeneous polynomial of bidegree b is a sum of monomials of bidegree b , so it is a linear combination of the corresponding basis monomials. Uniqueness follows from independence of monomials. $\square$

Thus one may represent a homogeneous element of bidegree b by its coordinate column vector in the absolute basis. This is how the implementation passes from symbolic homogeneous polynomials to exact linear algebra over the base domain. The remaining question of how differential data is supplied and partially inferred will be taken up in Section 5, after the necessary Smith-normal-form machinery has been established.

4 Smith Normal Form and PID Module Computations

With the coordinate language fixed, the next step is to collect the Smith-normal-form facts that drive the exact algebra in every bidegree.

Smith normal form

We first recall the normal-form theorem itself and the notation it provides.

Theorem 4.1 (Smith normal form). *Let R be a PID and $A \in M_{m \times n}(R)$. Then there exist invertible matrices U and V over R such that*

$$UAV = D$$

with D diagonal. The diagonal entries are unique up to multiplication by units and may be chosen so that each nonzero diagonal entry divides the next.

Theorem 4.1 will be used as a standard background result. It is the algebraic foundation of all later decomposition algorithms. By standard algorithms for Smith and Hermite normal forms over $\mathbb{Z}$ [8, 9], these normal-form computations can be carried out in polynomial time in the matrix dimensions and the bit-size of the input entries.

Solving linear equations

A second crucial use of Smith normal form is exact solvability of linear systems over a PID.

Proposition 4.2 (Linear solvability). *Let $A \in M_{m \times n}(R)$ and $T \in M_{m \times s}(R)$, where R is a PID. Suppose*

$$UAV = D$$

is a Smith normal form of A . Then the equation

$$AX = T$$

is solvable if and only if the transformed right-hand side UT satisfies the following row-by-row conditions:

1. if $1 \leq i \leq \min(m, n)$ and $D_{ii} \neq 0$, then the i -th row of UT is divisible by D_{ii} ,
2. if the i -th row of D is zero, then the i -th row of UT also vanishes.

Proof. Write $A = U^{-1}DV^{-1}$. Then

$$AX = T \iff DV^{-1}X = UT.$$

If $Y = V^{-1}X$, this becomes

$$DY = UT.$$

Since D is diagonal, this equation splits row by row. If $1 \leq i \leq \min(m, n)$ and $D_{ii} \neq 0$, one must have exact divisibility of the i -th row of UT by D_{ii} . If the i -th row of D is zero, then the corresponding row of DY is automatically zero, so the i -th row of UT must vanish. This includes both the case $D_{ii} = 0$ inside the diagonal range and the extra zero rows that occur when $m > n$. Conversely, if these conditions hold, one constructs Y row by row and then sets $X = VY$. $\square$

A particularly useful application in our spectral sequence computation is what we will call *alignment*.

An alignment lemma

Proposition 4.3 (Alignment). *Let $A \in M_{m \times k}(R)$ and $B \in M_{m \times n}(R)$, with R a PID. Assume the columns of A lie in the column space of B . Then there exist invertible matrices $P \in GL_k(R)$, $Q \in GL_n(R)$, and a diagonal matrix $D \in M_{n \times k}(R)$ such that*

$$AP = BQD.$$

This is called alignment because P and Q align A and B so that AP can be obtained by scaling columns of BQ .

Proof. Since the columns of A lie in the column space of B , there exists a matrix $X \in M_{n \times k}(R)$ such that

$$A = BX.$$

Take a Smith normal form of X :

$$UXV = D.$$

Then

$$X = U^{-1}DV^{-1}.$$

Hence

$$A = BU^{-1}DV^{-1}.$$

Multiplying on the right by V gives

$$AV = BU^{-1}D.$$

So we may take

$$P = V, \quad Q = U^{-1}.$$

$\square$

Structure extraction by alignment

The most useful invariant of a finitely generated module over a PID is its decomposition into free and torsion parts. In the present setting, one must first replace a raw span matrix by a basis matrix for its column space, and only then align the relations against that basis.

Definition 4.4. Let M be an R -module. A finite list $b_1, \dots, b_r \in M$ is a basis of M if the map

$$R^r \longrightarrow M, \quad (c_1, \dots, c_r) \longmapsto \sum_{i=1}^r c_i b_i$$

is an isomorphism of R -modules. Equivalently, every element of M can be written uniquely as an R -linear combination of $b_1, \dots, b_r$.

Proposition 4.5 (Basis extracted from a span matrix). *Let $S \in M_{N \times n}(R)$, and choose a Smith normal form*

$$USV = D.$$

Let r be the number of nonzero diagonal entries of D , and let B be the matrix consisting of the first r columns of SV . Then the columns of B form a basis of $\text{Col}(S)$.

Proof. Since V is invertible, S and SV have the same column space. But $SV = U^{-1}D$, so the first r columns of SV are the images under the invertible map U^{-1} of the nonzero diagonal columns of D . Those diagonal columns are visibly linearly independent, and they span the column space of D . Hence the first r columns of SV form a basis of $\text{Col}(S)$. $\square$

Proposition 4.3 now yields the following concrete structure result.

Proposition 4.6 (Aligned structure decomposition). *Let R be a PID, let $S \in M_{N \times r}(R)$ be a matrix whose columns form a basis of $\text{Col}(S) \subseteq R^N$, and let $Z \in M_{N \times k}(R)$ be a relation matrix whose columns lie in $\text{Col}(S)$. Choose invertible matrices $P \in GL_k(R)$, $Q \in GL_r(R)$, and a diagonal matrix D such that*

$$ZP = SQD.$$

Write the columns of SQ as $h_1, \dots, h_r$. Then

$$\text{Col}(S) / \text{Col}(Z) \cong \bigoplus_{i=1}^r R / (D_{ii}),$$

with the conventions:

1. *if $D_{ii} = 0$, then $R / (D_{ii}) = R$,*
2. *if D_{ii} is a unit, then $R / (D_{ii}) = 0$,*
3. *if i is outside the diagonal range of D , then $D_{ii} = 0$.*

Proof. This is where the hypothesis that the columns of S form a basis is used. Since Q is invertible, replacing S by SQ is a change of basis in the free module $\text{Col}(S)$. Therefore the columns $h_1, \dots, h_r$ also form a basis of $\text{Col}(S)$. By the definition of basis, every element of $\text{Col}(S)$ is written uniquely as a sum $\sum_i c_i h_i$, so

$$\text{Col}(S) = \bigoplus_{i=1}^r R h_i.$$

(Here we use the standard convention that R -modules are left modules.)

Since P is invertible, the columns of ZP generate the same relation submodule as the columns of Z . But

$$ZP = SQD,$$

so the relation submodule is generated by

$$D_{11}h_1, \dots, D_{\ell\ell}h_\ell,$$

where ℓ is the number of diagonal positions of D . Because $h_1, \dots, h_r$ are a basis, these generators impose independent cyclic relations in the direct-sum coordinates: the i -th relation lies only in the summand Rh_i . This independence is exactly what can fail if S is only a column-generating matrix rather than a basis matrix.

Thus

$$\text{Col}(S)/\text{Col}(Z) \cong \frac{\bigoplus_{i=1}^r Rh_i}{\bigoplus_{i=1}^\ell R(D_{ii}h_i)} \cong \bigoplus_{i=1}^r R/(D_{ii}),$$

with the stated conventions. □

Proposition 4.6 is precisely why Smith normal form is so natural here: after reducing a span matrix to a basis matrix for its column space, it converts a messy quotient of one column space by another into a direct sum decomposition. This is important for our program because after the homology computation, we need to extract structural information as our output.

Kernels

The same normal-form computation also gives an explicit description of kernels.

Proposition 4.7 (Kernel from Smith form). *Let $A \in M_{m \times n}(R)$ with Smith normal form*

$$UAV = D.$$

Then a basis for $\ker(A)$ is given by the columns of V corresponding to the free columns of D , namely:

1. *columns j with $j > m$, and*
2. *columns $j \leq \min(m, n)$ for which $D_{jj} = 0$.*

Proof. Since $A = U^{-1}DV^{-1}$, one has

$$Ax = 0 \iff DV^{-1}x = 0.$$

Thus $x \in \ker(A)$ if and only if $V^{-1}x \in \ker(D)$. But the kernel of a diagonal matrix is obvious: it is generated by the standard basis vectors in the free columns. Applying V carries those kernel generators back to generators of $\ker(A)$. □

This is exactly the algebraic content of computing kernels by SNF.

With the abstract recursion from Section 3 and the Smith-normal-form tools above in hand, we can now describe one of the main computations needed for page construction: preimages of submodules specified by column data.

Preimages of submodules given by column data

Proposition 3.2 shows that to pass from page E_r to E_{r+1} , one must compute

$$T_{r+1}^b = (\bar{d}_r)^{-1}(N_r^{b+\delta})$$

inside the ambient module F^b . Thus the relevant linear-algebra problem is the following: given a linear map between free modules and a matrix whose columns generate a submodule of the codomain, compute the preimage of this submodule. For future reuse, we shall generalize this problem to affine case.

Proposition 4.8 (Preimage block criterion). *Consider two matrices $S \in M_{n \times s}(R)$ and $K \in M_{m \times k}(R)$. Let $L: \text{Col}(S) \rightarrow R^m$ be a linear map. Then, for $v \in R^s$ and $y \in R^m$,*

$$L(Sv) \in y + \text{Col}(K) \iff \exists u \in R^k, \begin{bmatrix} L(S) & K \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix} = y$$

Therefore, $L^{-1}(y + \text{Col}(K))$ is obtained by the following affine linear computation:

1. solve

$$\begin{bmatrix} L(S) & K \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix} = y$$

using Proposition 4.2;

2. if no solution exists, then $L^{-1}(y + \text{Col}(K))$ is empty;

3. if (v_0, u_0) is one solution, and if

$$\begin{bmatrix} V_{\ker} \\ U_{\ker} \end{bmatrix}$$

is a matrix whose columns form a basis of the kernel of $\begin{bmatrix} L(S) & K \end{bmatrix}$, then all solutions are obtained by adding its column space, using Proposition 4.7;

4. project this affine solution set to the first s coordinates, obtaining $v_0 + \text{Col}(V_{\ker})$;

5. then

$$L^{-1}(y + \text{Col}(K)) = Sv_0 + \text{Col}(SV_{\ker}).$$

Proof. An element of $\text{Col}(S)$ has the form Sv for some coefficient vector v . By definition,

$$Sv \in L^{-1}(y + \text{Col}(K)) \iff L(Sv) \in y + \text{Col}(K) \iff L(S)v - y \in \text{Col}(K).$$

The last condition holds if and only if there exists $w \in R^k$ such that

$$L(S)v - y = Kw.$$

Equivalently, after setting $u = -w$, this becomes

$$\begin{bmatrix} L(S) & K \end{bmatrix} \begin{bmatrix} v \\ u \end{bmatrix} = y.$$

This proves the displayed equivalence. If a particular solution (v_0, u_0) exists, all solutions are obtained by adding elements of the kernel of the block matrix $\begin{bmatrix} L(S) & K \end{bmatrix}$. Projecting this affine solution set to the first s coordinates gives exactly the possible preimage coefficient vectors v . Multiplying those vectors by S converts them into elements of $\text{Col}(S)$, so the full preimage is the affine translate $Sv_0 + \text{Col}(SV_{\ker})$. $\square$

Proposition 4.8 reduces affine preimage computations to one exact linear solve together with the kernel of the same block matrix. Combined with the image-column-space computations above, this is enough for the homology step. However, we still need to fill in the last theoretical piece.

Division of Homogeneous Elements and Uniqueness of Quotients

Let $x \in E_r^\alpha$ be homogeneous, and let $y \in E_r^{b+\alpha}$ be another homogeneous element. We ask whether there exists $q \in E_r^b$ such that

$$xq = y$$

in the quotient module $E_r^{b+\alpha}$, and if so whether this quotient class is unique. This exact division problem is the basic algebraic subroutine behind the reverse-Leibniz and alignment steps used to recover unknown differential values.

Write the relevant homogeneous pieces in the standard column-space presentation

$$E_r^\alpha \cong \text{Col}(S_\alpha) / \text{Col}(Z_\alpha), \quad E_r^b \cong \text{Col}(S_b) / \text{Col}(Z_b), \quad E_r^{b+\alpha} \cong \text{Col}(S_{b+\alpha}) / \text{Col}(Z_{b+\alpha}),$$

with all vectors written in absolute coordinates. Choose a representative $\tilde{x} \in \text{Col}(S_\alpha)$ of the class x . Multiplication by x sends E_r^b to $E_r^{b+\alpha}$; at the representative level, let $\tilde{x}S_b$ denote the matrix whose columns are the products of $\tilde{x}$ with the corresponding columns of S_b . Let $\tilde{y} \in F^{b+\alpha}$ be any representative of the class y .

Proposition 4.9 (Exact division criterion). *There exists $q \in E_r^b$ such that $xq = y$ in $E_r^{b+\alpha}$ if and only if the representative $\tilde{y}$ lies in the column space of*

$$[\tilde{x}S_b \mid Z_{b+\alpha}].$$

Equivalently, the division problem is solvable if and only if the exact linear system

$$[\tilde{x}S_b \mid Z_{b+\alpha}] \begin{bmatrix} v \\ u \end{bmatrix} = \tilde{y}$$

is solvable. This criterion is independent of the chosen representatives $\tilde{x}$ and $\tilde{y}$.

Proof. Apply Proposition 4.8 with $S = S_b$, $K = Z_{b+\alpha}$, and $L: \text{Col}(S_b) \rightarrow F^{b+\alpha}$ given by multiplication by the chosen representative $\tilde{x}$. Then $L(S_b)$ is exactly the matrix $\tilde{x}S_b$. A representative $q = S_b v$ satisfies $xq = y$ in $E_r^{b+\alpha}$ exactly when

$$L(S_b v) \in \tilde{y} + \text{Col}(Z_{b+\alpha}).$$

By Proposition 4.8, this is equivalent to the existence of u such that

$$[\tilde{x}S_b \mid Z_{b+\alpha}] \begin{bmatrix} v \\ u \end{bmatrix} = \tilde{y}.$$

This is precisely the condition that $\tilde{y}$ lie in the column space of the displayed block matrix. If $\tilde{y}'$ is another representative of y , then $\tilde{y}' - \tilde{y} \in \text{Col}(Z_{b+\alpha})$, so solvability for $\tilde{y}$ is equivalent to solvability for $\tilde{y}'$ by absorbing that difference into the u -coordinate.

It remains to check that the criterion does not depend on the representative of x . If $\tilde{x}'$ is another representative of x , then $\tilde{x}' - \tilde{x} \in \text{Col}(Z_\alpha)$. Since multiplication is well defined on the quotient page, every product of an element of $\text{Col}(Z_\alpha)$ with an element of $\text{Col}(S_b)$ lies in $\text{Col}(Z_{b+\alpha})$. Therefore the corresponding columns of $\tilde{x}'S_b - \tilde{x}S_b$ lie in $\text{Col}(Z_{b+\alpha})$. It follows that

$$\text{Col}([\tilde{x}'S_b \mid Z_{b+\alpha}]) = \text{Col}([\tilde{x}S_b \mid Z_{b+\alpha}]),$$

so the solvability condition is independent of the representative chosen for x . $\square$

When the division problem is solvable, Proposition 4.8 also describes the full affine solution set. Choose one solution (v_0, u_0) , and let K_{raw} be the projection of the kernel of

$$[\tilde{x}S_b \mid Z_{b+\alpha}]$$

to the v -coordinate rows. Then $q_0 = S_b v_0$ is one quotient representative, and the possible changes in the representative are generated in absolute coordinates by the columns of $S_b K_{\text{raw}}$. Thus the only part not already covered by Proposition 4.8 is the passage to the quotient in which the unknown q lives:

$$E_r^b = \text{Col}(S_b) / \text{Col}(Z_b).$$

The relevant ambiguity is the image of $\text{Col}(S_b K_{\text{raw}})$ in this quotient, namely

$$\frac{\text{Col}(S_b K_{\text{raw}}) + \text{Col}(Z_b)}{\text{Col}(Z_b)} \cong \frac{\text{Col}(S_b K_{\text{raw}})}{\text{Col}(S_b K_{\text{raw}}) \cap \text{Col}(Z_b)}.$$

We call this quotient the *division ambiguity module*.

Proposition 4.10 (Uniqueness of exact division). *The quotient q is unique in E_r^b if and only if the division ambiguity module is zero. Equivalently,*

$$\text{Col}(S_b K_{\text{raw}}) \subseteq \text{Col}(Z_b).$$

Proof. Two representatives $q_0 + k_1$ and $q_0 + k_2$ determine the same class in E_r^b if and only if their difference $k_1 - k_2$ lies in $\text{Col}(Z_b)$. Hence all representatives define the same quotient class exactly when every ambiguity vector coming from $\text{Col}(S_b K_{\text{raw}})$ is killed by the relations in E_r^b . $\square$

This kind of uniqueness is crucial for the differential computations in Section 5. For uniqueness alone, we do not need the full structure of the division ambiguity module, which would require the basis extraction step from Proposition 4.5. It is enough to test the inclusion using Proposition 4.2

$$\text{Col}(S_b K_{\text{raw}}) \subseteq \text{Col}(Z_b).$$

5 Computing Differential Images

Supplying differential data

We now return to the third practical question raised in Section 3: how is differential information supplied to the computation?

A naive interface would ask the user to supply differential values on a spanning set of T_r^b . In practice this is usually unnecessary. For example, some values are automatic from the quotient presentation itself: elements of N_r^b already represent zero on page r , so their differential must also be zero on that page.

A slightly more involved case is that we may know $d(J)$ for a matrix J whose columns span only part of T_r^b , and whose generators are quite different from the span matrix S_b currently stored by the program. In that situation, one wants to use as much of this partial information as possible by aligning J with S_b , and then addressing any remaining unresolved directions separately.

The most delicate source of additional information is the Leibniz rule. In particular, exact division makes it possible to run reverse-Leibniz arguments, where one tries to recover $d(q)$ from $d(xq)$ and $d(x)$.

The following subsections isolate these ingredients and then assemble them into the per-bidegree completion strategy used by the implementation.

Known information and alignment

We first discuss the alignment step. Fix a source bidegree b on page r . Let

$$S = \begin{bmatrix} s_1 & \cdots & s_n \end{bmatrix}$$

be a span matrix for the module E_r^b . Suppose also that we know the differential on a matrix

$$J = \begin{bmatrix} x_1 & \cdots & x_k \end{bmatrix}$$

of source elements at that same bidegree, together with their images

$$d(J) = \begin{bmatrix} d(x_1) & \cdots & d(x_k) \end{bmatrix}.$$

In the implementation, J is assembled from the currently known source elements at bidegree b , together with any source relations that automatically have zero differential.

If the columns of J lie in the column space of S , then by Proposition 4.3 there exist invertible matrices P, Q and a diagonal matrix D such that

$$JP = SQD.$$

Applying the differential gives

$$d(J)P = d(S)QD.$$

This equation is extremely useful because the right-hand side is diagonal with respect to the transformed source generators SQ . Then one reduces the problem of determining $d(S)$ to a collection of exact division problems:

1. for each nonzero diagonal entry D_{ii} , solve

$$D_{ii}y_i = \rho_i,$$

with uniqueness understood modulo the relation submodule in the differential target bidegree,

2. if $D_{ii} = 0$, then one must have $\rho_i = 0$; otherwise the known differential data is inconsistent,
3. any zero columns of $d(J)P$ beyond the diagonal part must also vanish.

Any transformed source columns that remain ambiguous after this process are then passed to the recursive Leibniz and exact-division inference routine discussed below. Only the columns still unresolved after that stage require extra user input.

Reverse Leibniz inference

Suppose x and q are homogeneous elements on page r . Then

$$d_r(xq) = d_r(x)q + x d_r(q).$$

If x and q are already known classes on the page, and if $d_r(x)$ and $d_r(xq)$ are known, then this equation may determine $d_r(q)$ by an exact division problem whose unknown lies in the same bidegree as $d_r(q)$. In the implementation, one often reaches this situation by first recovering a quotient representative from a known product xq , but the proposition below isolates the exact-division step once the classes x and q are fixed.

Proposition 5.1 records a useful sufficient criterion for carrying this out.

Proposition 5.1 (Reverse Leibniz inference). *Let x, q be homogeneous elements on a fixed page. Let δ be the bidegree of the differential on that page, and write $|q| = \beta$. Assume:*

1. $d(x)$ and $d(xq)$ are known,
2. the equation

$$xz = d(xq) - d(x)q$$

has a unique solution z as a class in $E_r^{\beta+\delta}$.

Then $d(q)$ is uniquely determined in $E_r^{\beta+\delta}$ by

$$d(q) = z.$$

Proof. This is just the Leibniz identity rewritten as

$$x d(q) = d(xq) - d(x)q.$$

Since x and q are already given as page classes, the right-hand side is a well-defined class in the product bidegree. The second hypothesis says that the resulting division by x has a unique solution class in $E_r^{\beta+\delta}$. Since $d(q)$ is one such solution by the Leibniz identity, it must equal z . $\square$

Implementation of Leibniz inference

The proposition above is a clean sufficient criterion, but the implementation uses a more flexible inference mechanism. Rather than insisting that the quotient q be unique before any new data are stored, the code works with a differential knowledge base together with a distinguished set of *lowest* known sources. Conceptually, the lowest set consists of the currently known nontrivial sources that are not yet recognized as products of other known sources.

When a new differential pair $y \mapsto d(y)$ is added, the implementation first applies constant-coefficient division when possible. If $y = cq$ for a nonunit scalar c , and if division by c is unique in both source and target, then the code also stores the inferred pair

$$q \mapsto d(y)/c.$$

The implementation then tries reverse-Leibniz factorizations against the current lowest set. Suppose x is a known source with known differential and that division of y by x is solvable. Choose one quotient representative q_0 , and let $g_1, \dots, g_n$ be generators for the division ambiguity module, so that every quotient representative has the form

$$q_0 + \sum_i c_i g_i.$$

The code attempts to solve

$$x d(q_0) = d(y) - d(x)q_0$$

and, for each ambiguity generator g_i ,

$$x d(g_i) = -d(x)g_i.$$

In the formulation used here, any uniquely determined solutions obtained from these equations are stored immediately in the knowledge base, even if the entire ambiguity family is not resolved at once. Successful additions are then allowed to update the lowest set.

For an unknown source s , the recursive query routine searches over left factors x in the lowest set, tries to divide s by each such x , and recursively asks for the differentials of the resulting quotient representative and ambiguity generators. If these recursive calls succeed and all ambiguity choices give the same class modulo relations in the differential target bidegree, then the inferred differential $d(s)$ is stored as a new known value.

Thus the implemented Leibniz mechanism is not a full forward derivation closure. Instead, it is a controlled search procedure based on exact division, reverse-Leibniz equations, ambiguity checks, and incremental enlargement of the knowledge base.

The differential-completion strategy

At a fixed source bidegree b , the overall strategy is as follows.

1. Gather all currently known differential pairs at bidegree b , together with any source relations whose differential is automatically zero. During this step, the lowest set is built.
2. Before any alignment step, query the differential value on the span matrix S against the lowest set. (This process also potentially updates the lowest set in the way described above.)
3. Form the known-data matrix J from the information currently available at bidegree b , and align it with the full source span matrix S , obtaining

$$JP = SQD.$$

4. Apply the differential and solve

$$d(J)P = d(S)QD$$

column by column. This determines all directly known transformed source columns and any additional columns for which the resulting exact division is unique modulo the relations in the differential target bidegree.

5. If some transformed source columns remain unresolved, attempt recursive Leibniz and exact-division inference on those transformed generators.
6. If ambiguity still remains, request the missing differential values from the user.
7. Once all transformed columns are known modulo the relations in the differential target bidegree, convert back to the original source generators.

What matters mathematically is that this procedure reduces the global problem of finding a differential image column space to exact division problems over a PID, together with a controlled notion of quotient ambiguity.

6 Some Computation Details

This section collects a few additional computational remarks that are useful in practice.

Page numbering in the implementation

The paper uses the standard mathematical notation $E_1, E_2, E_3, \dots$. The implementation follows the same convention, but this is worth stating explicitly because program data structures often use zero-based indexing.

Internally, the spectral-sequence object stores a dummy entry in position 0, so that the page object at index r is exactly page E_r . Consequently, the first call to `add_page` constructs E_1 and equips it with the differential d_1 , the second call constructs E_2 , and so on. In other words, page number and construction order agree.

This convention matches the notation used throughout the paper, but it also means that the implementation has no separate page E_0 object. All examples and code snippets should therefore be read with

$$p1 = \text{ss.add_page}(\dots) \iff p1 = E_1.$$

In some cases, the mathematical setup starts with a description of E_2 rather than E_1 . In that situation, one should still construct a trivial E_1 with no differential to keep the page indexing correct. This matters because the page number r affects the bidegree of d_r .

Computational PID domains

The abstract results above only require a PID. The actual algorithms, however, require exact effective operations. In particular, the implementation uses:

1. extended gcd,
2. remainders,
3. exact quotients by divisors.

Accordingly, one should distinguish the following notion.

The next definition isolates exactly that extra structure.

Definition 6.1. A *computational PID domain* is a PID equipped with exact algorithms for the operations required by the SNF procedures above.

In the current implementation, we use the Smith-normal-form routines provided by SymPy's exact domain-matrix layer [10], which rely on the operations listed above. More importantly, the higher-level page, kernel, alignment, and quotient computations are written in terms of this exact linear-algebra backend. Thus, for another exact coefficient domain with the same underlying operations, one would mainly need to replace that backend rather than redesign the surrounding spectral-sequence logic.

7 A Program Guide over $\mathbb{Z}$

This section explains how to use the program on the sample spectral sequence. The example we take is the Serre spectral sequence of a fibration over S^3 [7, 1, 6], but here we use only its algebraic description.

Algebraic description

The input algebra is

$$\mathbb{Z}[a, t]/(a^2),$$

with generator bidegrees

$$|a| = (3, 0), \quad |t| = (0, 2).$$

We work in the first quadrant, so all modules outside it are zero. The differential on page r has bidegree

$$(r, 1 - r).$$

The page data entered into the program are:

1. $d_1(a) = 0$ and $d_1(t) = 0$,
2. $d_2(a) = 0$ and $d_2(t) = 0$,
3. $d_3(t) = a$.

In the current sample driver `src/main.py`, one additional zero datum is supplied on page 3:

$$d_3(a) = 0.$$

This is redundant mathematically, because $d_3(a) = d_3(d_3(t)) = 0$, but we haven't integrated this type of inference in the current version.

Constructing the pages

The object is created by specifying the base domain, the generators, their bidegrees, and the coefficient matrix that produces the differential bidegree $(r, 1 - r)$:

```
from sympy import ZZ
from sympy.abc import a, t
from src.spectral_sequence import SpectralSequence

ss = SpectralSequence(
    ZZ,
    [a, t],
    [[3, 0], [0, 2]],
    [[1, 0], [-1, 1]],
)
```

The algebra relation is added by

```
ss.kill(a**2)
```

and the first four pages are then constructed by

```
p1 = ss.add_page({a: 0, t: 0})
p2 = ss.add_page({a: 0, t: 0})
p3 = ss.add_page({t: a, a: 0})
p4 = ss.add_page()
```

Because the implementation numbers pages literally, the first call to `add_page` creates E_1 , the second creates E_2 , and the third creates E_3 . So the d_3 -data are supplied in the third call.

Inspecting later pages

After a page has been built, the module at a given bidegree is accessed by indexing the page object:

```
module = p4[3, 2]
info = module.get_structural_information()
```

This returns structural information for $E_4^{3,2}$. Likewise, one may inspect

$$E_4^{3,4}, \quad E_4^{3,6}, \quad E_4^{3,8}, \dots$$

by querying the corresponding bidegrees on $p4$.

In this sample, the program reports torsion of orders

$$2, \quad 3, \quad 4, \quad 5, \dots$$

at those bidegrees. The point of this section is not to rederive that pattern manually, but to show how it is recovered by the quotient-module and Smith-normal-form routines developed above.

Generating a chart

The sample driver `src/main.py` also exports an HTML chart. From the repository root, one runs

```
python src/main.py <min_x> <max_x> <min_y> <max_y> <out.html>
```

For example,

```
python src/main.py 0 8 0 8 output_current_main.html
```

builds the sample spectral sequence, scans the specified rectangle of bidegrees, and writes the resulting chart to an HTML file. The following is a screenshot of its output.

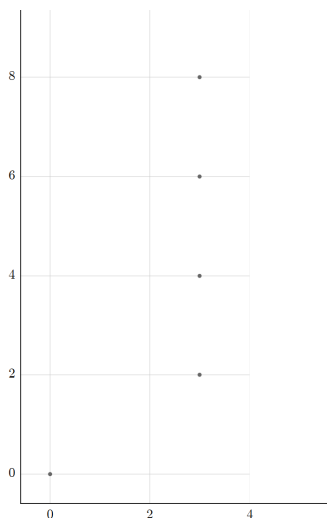


Figure 1: Screenshot of the generated HTML chart for the sample spectral sequence.

What the guide illustrates

This sample shows how the abstract formalism of the paper appears in actual use:

1. the first page is specified by generators, bidegrees, and algebra relations,
2. page differentials are given as sparse dictionaries of known values,
3. later pages are produced automatically by homology computations in quotient modules,
4. structural information is extracted from those quotient modules by Smith normal form.

8 Implementation Scope and Limitations

Before concluding, we summarize the main hypotheses and limitations built into the present implementation.

The mathematical framework developed above is quite general, but the present implementation works under a more specific set of assumptions and still has several important limitations.

Assumptions

1. The base ring and the algebra are assumed commutative.
2. The implemented algorithms require a computational PID domain in the sense of Section 6, not an arbitrary abstract PID. In practice, the code is aimed at exact Euclidean-style domains such as $\mathbb{Z}$ and exact fields.
3. The implementation works in the first quadrant: modules $E_r^{p,q}$ with $p < 0$ or $q < 0$ are taken to be zero.

Limitations

1. The differential-inference code does not yet exploit every consequence of the available data. In particular, it does not systematically use all zero values forced by vanishing target bidegrees, nor every consequence of the relation $d_r^2 = 0$.
2. The implementation does not compute the full Leibniz closure of the supplied differential data, so some differential values that are mathematically determined may still be missed.
3. If differential data remain ambiguous, the current program asks for additional user input rather than continuing with a formal treatment of undetermined classes. One could imagine extending the implementation to track such unresolved data explicitly, but that is not done here.
4. The framework is intentionally algebraic and computational. It does not prove convergence or identify an abutment.

9 Conclusion

We close by restating the main conceptual point and indicating the broader significance of the method.

The main point of this paper is that the computation of algebraic spectral sequences over PID-type coefficients can be organized around a single structural idea: represent each homogeneous piece as a quotient of one explicit column space by another inside an ambient free module, and reduce the required exact linear algebra to Smith-normal-form calculations, exact solves, kernel computations, and alignment.

Once this point of view is adopted, several operations become conceptually unified:

1. extracting generators and torsion from a page module,
2. deciding whether an element is zero or nontrivial,
3. computing kernels in quotient modules,
4. computing incoming image submodules,
5. reconstructing differential image column spaces from partial information.

The program guide based on $\mathbb{Z}[a, t]/(a^2)$ shows that even a very small multiplicative spectral sequence can produce nontrivial torsion patterns on later pages, and that Smith normal form detects these patterns immediately.

There are several natural directions for future work. One could strengthen the differential inference system, extend the supported coefficient domains, or connect the algebraic computations more explicitly to topological examples. But even in its current form, the framework already gives a coherent answer to the basic computational question: given algebraic data for a spectral sequence over PID-type coefficients, how can one compute later page modules and reconstruct differential image data?

The answer developed here is: by reducing the problem, bidegree by bidegree, to exact module computations, together with supplied or uniquely inferred differential data, and solving those computations with Smith normal form.

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